

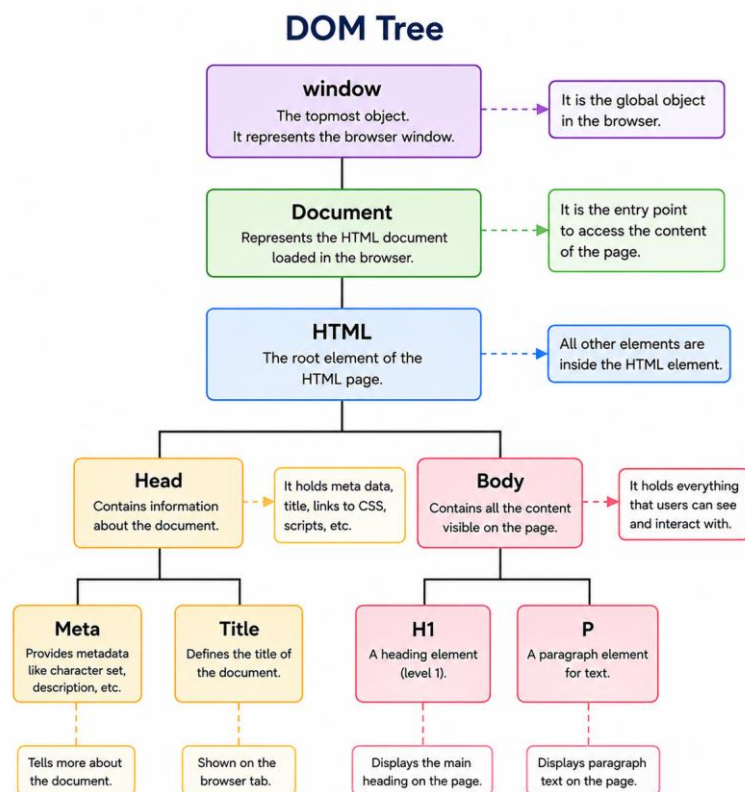
Module 55 - DOM Selecting Elements

1. What is DOM?

DOM (Document Object Model) is a programming interface that represents an HTML document as a tree of objects, allowing JavaScript to access and modify the HTML elements and content.

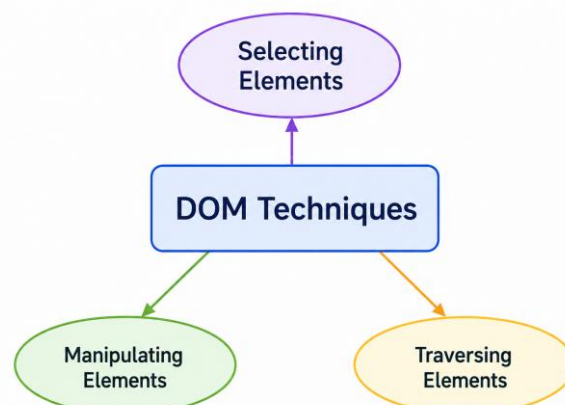
2. What is DOM Tree?

DOM Tree is a hierarchical structure that represents an HTML document as a tree of nodes, where each HTML element is connected as a parent, child, or sibling.



3. What are DOM Techniques?

DOM Techniques are methods used in JavaScript to select, manipulate, and traverse HTML elements in a webpage.



4. What is DOM Selecting Elements?

DOM Selecting Elements is the process of finding and accessing HTML elements from a webpage using JavaScript.

Common DOM Selection Methods

- **getElementsByTagName()** → Selects elements by their HTML tag name.
- **getElementsByClassName()** → Selects elements by their class name.
- **getElementById()** → Selects an element by its ID.
- **getElementsByName()** → Selects elements by their name attribute.
- **querySelector()** → Selects the first matching element using a CSS selector.
- **querySelectorAll()** → Selects all matching elements using a CSS selector.

Example Code:

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Selecting Elements</title>
</head>
<body>
  <h1>DOM - Document Object Model</h1>
  <h1>Selecting Elements</h1>
  <p class="para1">Content - 1</p>
  <p class="para1">Content - 2</p>
  <p id="uniq-para">Content - 3</p>
  <p id="uniq-para">Content - 4</p>
  <input type="text" name="input-1" placeholder="User Name">
  <script src="index.js"></script>
</body>
</html>
```

Index.js

```
// getElementsByTagName()
let heading = document.getElementsByTagName("h1");
console.log(heading);

// getElementsByClassName()
let para1 = document.getElementsByClassName("para1");
console.log(para1);

// getElementById()
let uniqPara = document.getElementById("uniq-para");
console.log(uniqPara);

// getElementsByName()
let nameAttribute = document.getElementsByName("input-1");
console.log(nameAttribute);

// querySelector()
// let selectOne = document.querySelector("h1");
// let selectOne = document.querySelector(".para1");
let selectOne = document.querySelector("#uniq-para");
// innerHTML gets the HTML content inside the selected element.
console.log(selectOne.innerHTML);

// querySelectorAll()
// let multiElements = document.querySelectorAll("h1");
// let multiElements = document.querySelectorAll(".para1");
let multiElements = document.querySelectorAll("#uniq-para");
console.log(multiElements);
```